ELT3207 Meas. & Digital Control Basics

Lecture 4: Data Acquisition Systems

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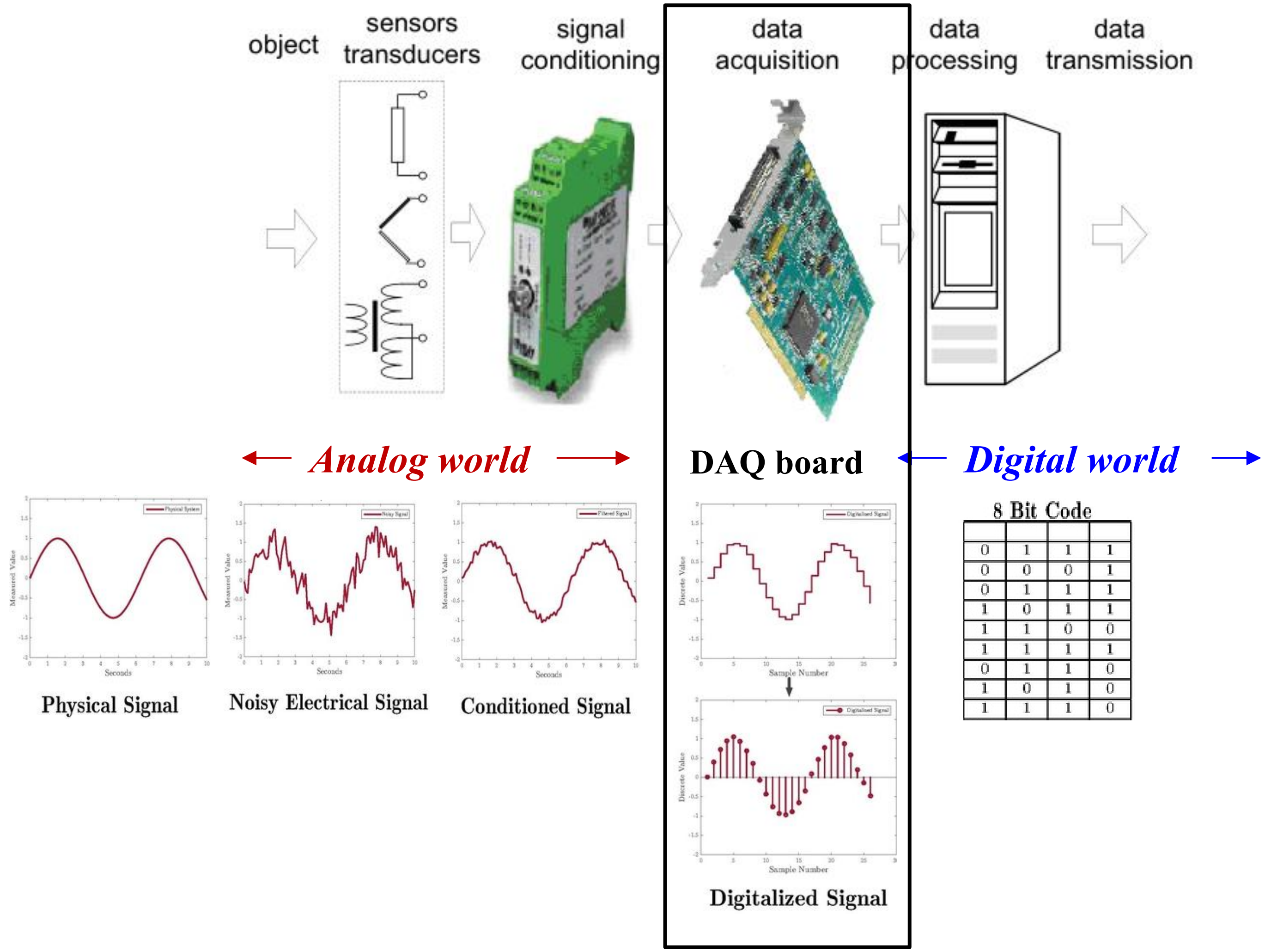
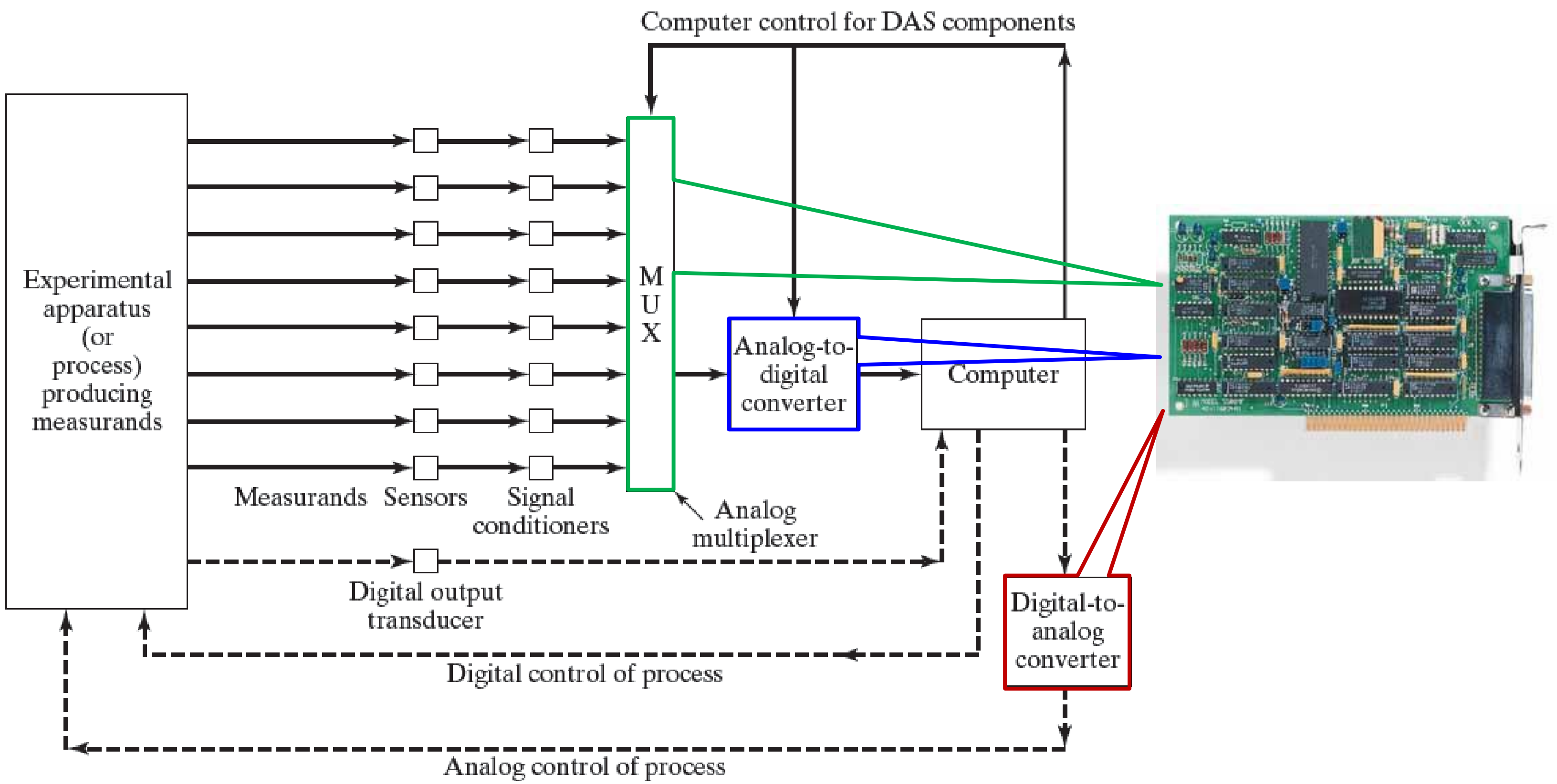
Introduction

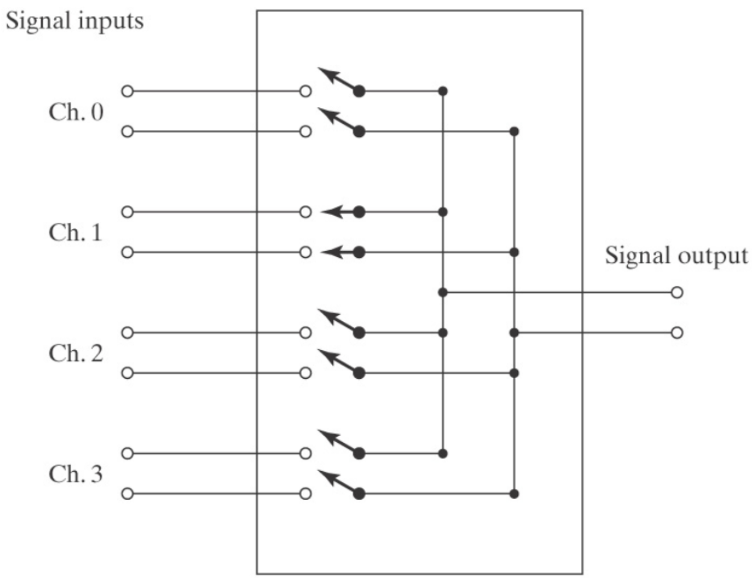


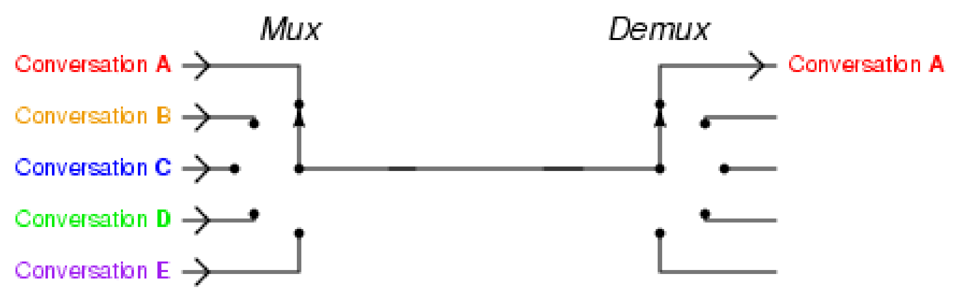
0	1	1	1
0	0	0	1
0	1	1	1
1	0	1	1
1	1	0	0
1	1	1	1
0	1	1	0
1	0	1	0
1	1	1	0

Schematic of a computerized measurement & control system



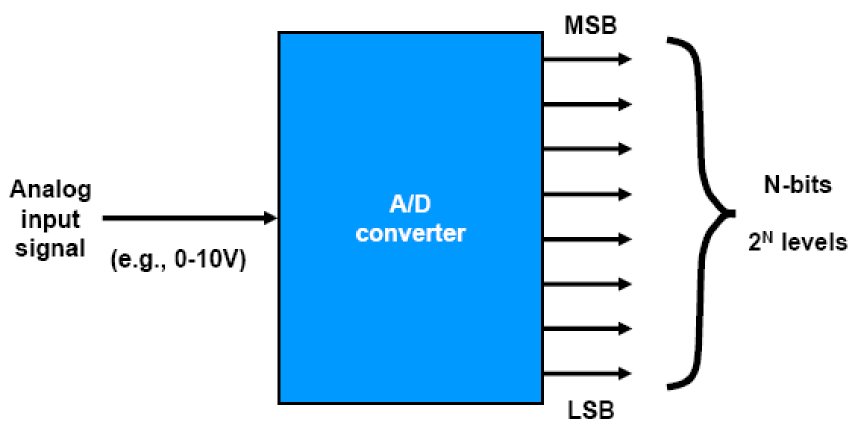
Multiplexer

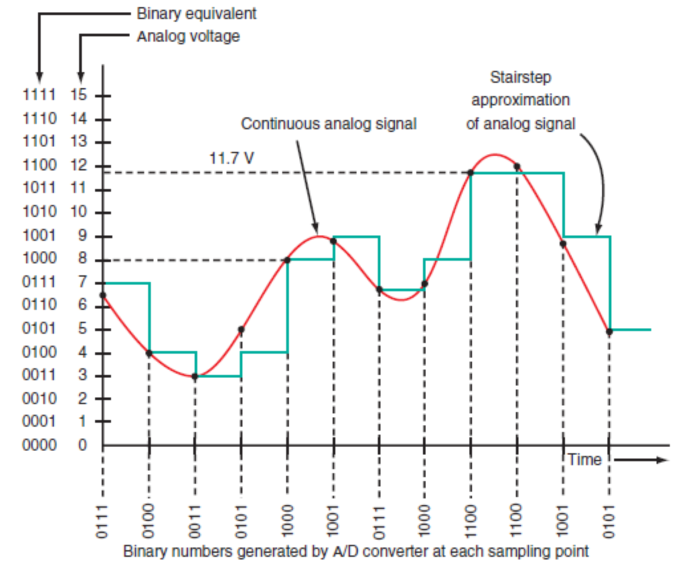




- Multiplexer (**MUX**): a device that selects one of many analog or digital data sources and outputs that source into a single channel.
- Demultiplexer (**DEMUX**) a device that that selects one of many data-output-lines and connects the selected output line to one single input.
  - A multiplexer is often used with a complementary demultiplexer on the receiving end.

Basics of Analog-to-Digital Conversion





- A/D: the process of turning an analog voltage into a digital number. The device performing this function is called an ADC.
  - The digital number represents the input voltage in discrete steps with finite **resolution**.
  - ADC resolution is determined by the number of bits that represent the digital number.
  - **Example**: an 8-bit converter with a full-scale voltage of 10V will give you a resolution of 10V/256 which is 39.1 mV.

ADC process

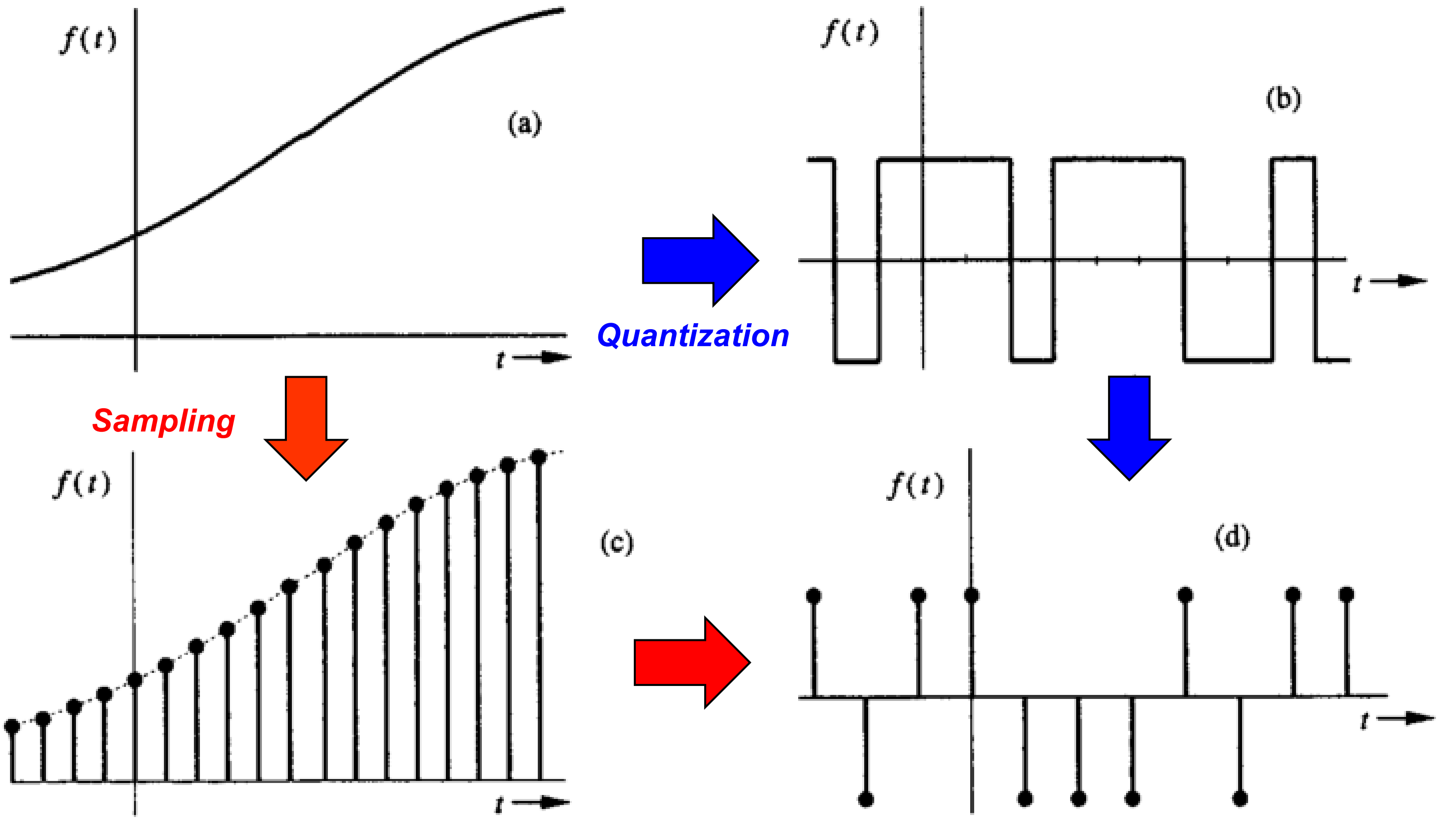
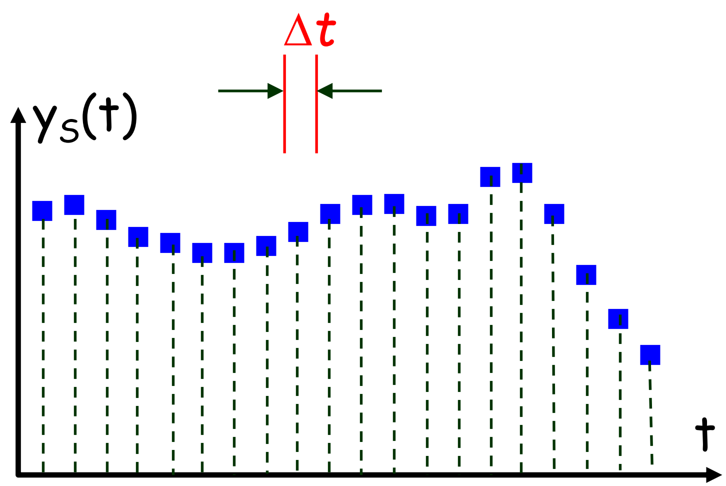
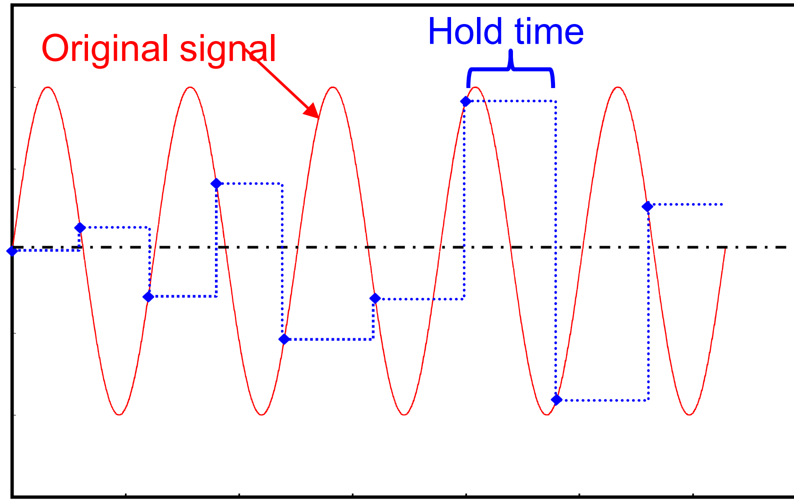


Fig. 1.5 Examples of Signals: (a) analog, continuous-time (b) digital, continuous-time (c) analog, discrete-time (d) digital, discrete-time.

- The terms continuous-time and discrete-time qualify the nature of signal along the time (horizontal) axis while the terms analog and digital qualify the nature of signal amplitude (vertical axis).

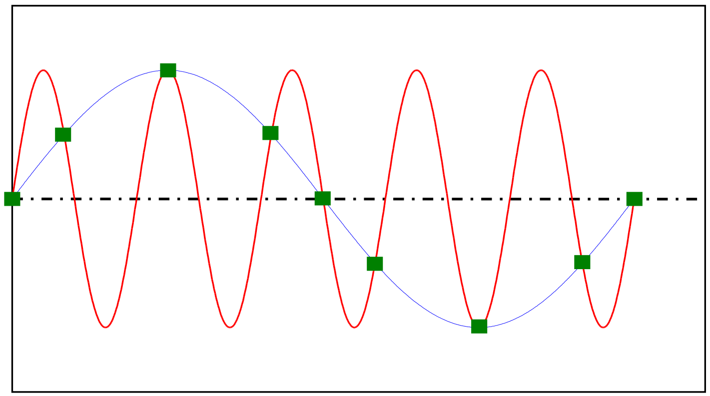
Sampling

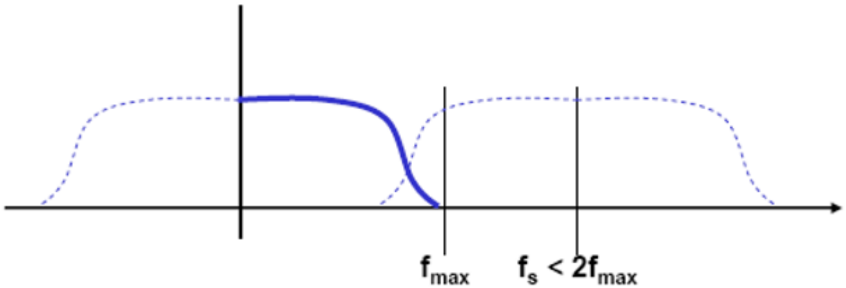




- Sampling interval $\Delta t \rightarrow$ sampling rate: $f_s = \frac{1}{\Delta t}$.
- Almost any analog to digital converter employ sample-and-hold technique, which holds each sample value (voltage) until the next pulse occurs.
  - The sampling and hold unit is necessary because the A/D converter requires a finite amount of time.

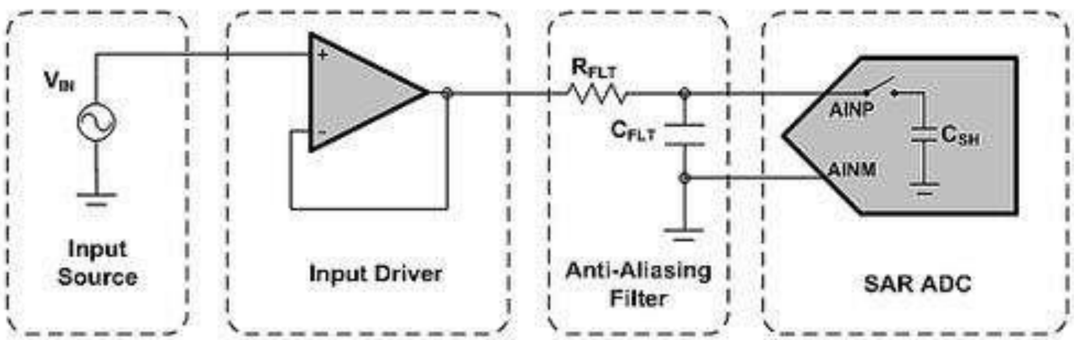
Aliasing and sampling theorem

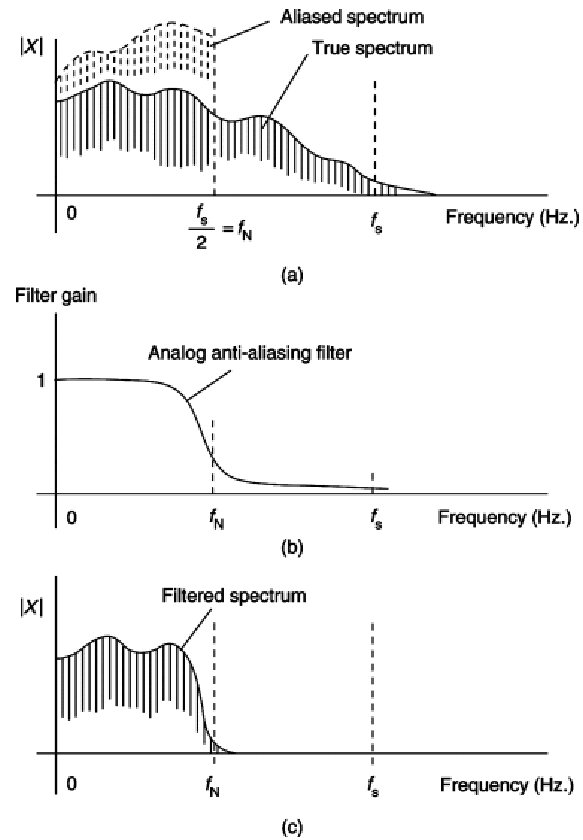




- **Aliasing**: an effect that causes different signals to become indistinguishable (or aliases of one another) when sampled.
- **Sampling theorem**: the sampling rate must be greater than twice the highest-frequency component of the original signal in order to reconstruct the original signal correctly, i.e. $f_s > 2f_m$, where f_m is the signal frequency (or the maximum signal frequency if there is more than one frequency in the signal).

Practical application of sampling theorem





- How does one avoid this alias phenomenon when sampling a signal of **unknown** frequency content?
  - ✓ Assign a sample rate so large that the measured signal does not have significant amplitude content above the Nyquist freq.
  - ✓ Pass the signal through a low-pass filter (**AAF**) based on the **maximum frequency of interest** prior to sampling.
- **Waveform Fidelity**: the sampling theorem ensures that there will be no aliasing associated with frequency content, it does not address the waveform distortion.
  - ✓ E.g., Super Audio CD (SACD) format recordings. As a **general rule** waveform fidelity is acceptable for $f_s \geq 5f_m$ to $10f_m$ depending on the application.

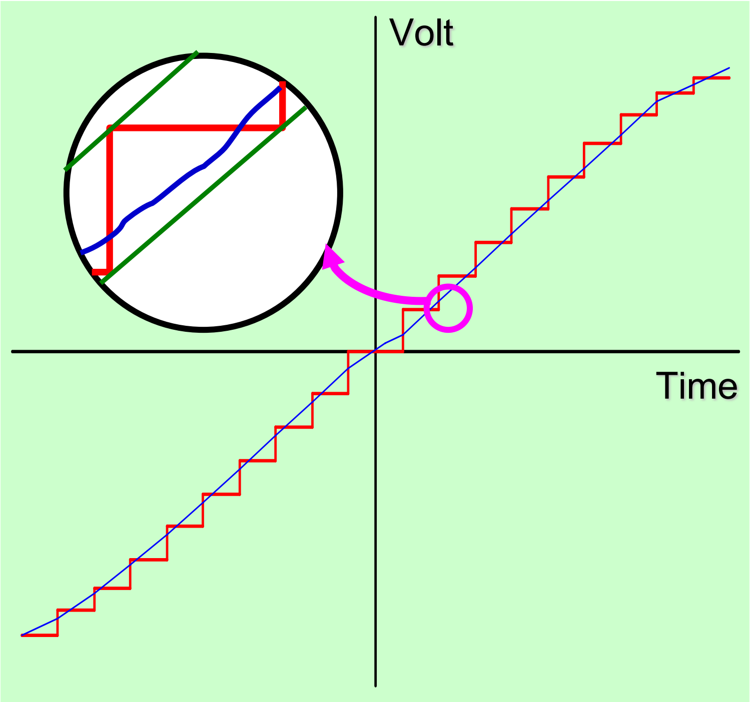
Quantization basics

- Quantization is the process to convert the input voltage range into $Q = 2^N$ bands to encode a continuous analog signal to discrete digital levels.
  - Quantization Interval:

    $\Delta V = \frac{V_{ru} - V_{rl}}{Q}$.
  - Quantization Error:

    $e_q^{\max} = \pm \frac{\Delta V}{2(V_{ru} - V_{rl})} \times 100\%$

    and is treated as an uncertainty, commonly referred to as **input resolution error**.
  - **Saturation error** occurs if $V_i > V_{ru}$ or $V_i < V_{rl}$ and is considered a mistake which should have been avoided.



Quantization output encoding

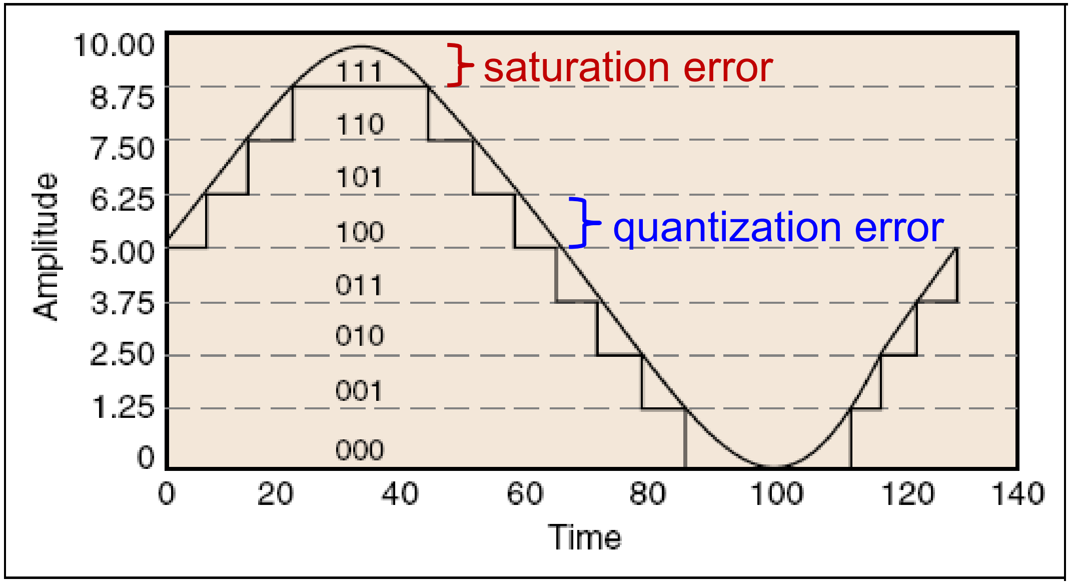


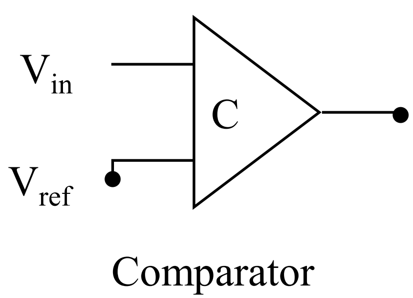
Figure 3. Digitized Sine Wave with a Resolution of Three Bits

- There are a number of ways to encode the digital output of ADCs
  - **Binary encoding**: N bits represent 2^N possible values for the output $\rightarrow N$ determines the resolution/quantization interval of the output: $LSB = \frac{V_{ru} - V_{rl}}{2^N}$.

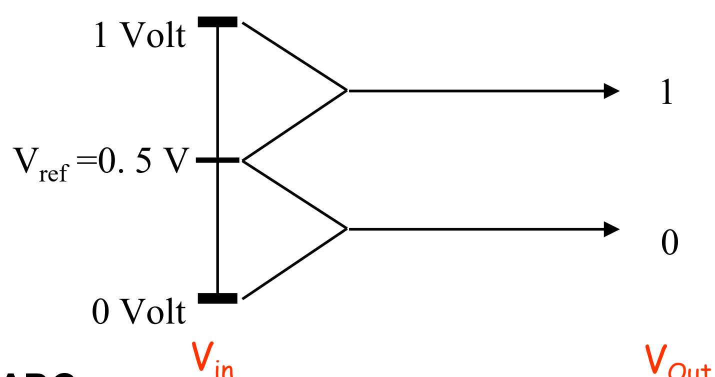
Bits	1	2	6	8	10	12
$2^N =$	2	4	64	256	1024	4096
Res. %	50	25	1.6	0.39	0.1	0.024

Quantization & binary encoding (cont.)

- **Input range**: can be unipolar or bipolar
  - A **unipolar** converter can only respond to analog inputs with the same sign
  - **Bipolar** converters can convert both positive and negative analog inputs
- In general, the output of an ADC has 2^N possible values, where N is the number of bits used to represent the digital output.
  - The 1-bit device has two possible output states (0,1); the 2-bit device has 4 possible output states (00, 01, 10, and 11 in binary representation).
  - A comparator is the most basic 1-bit ADC



$$\textbf{Out} = 0 \text{ if } V_{in} < V_{ref}$$
$$= 1 \text{ if } V_{in} > V_{ref}$$



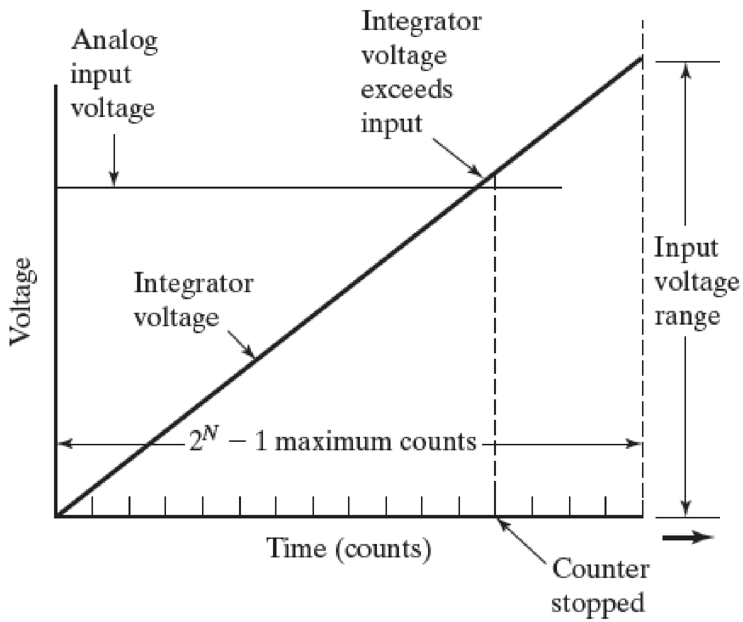
- **This is a 1-bit Flash ADC**

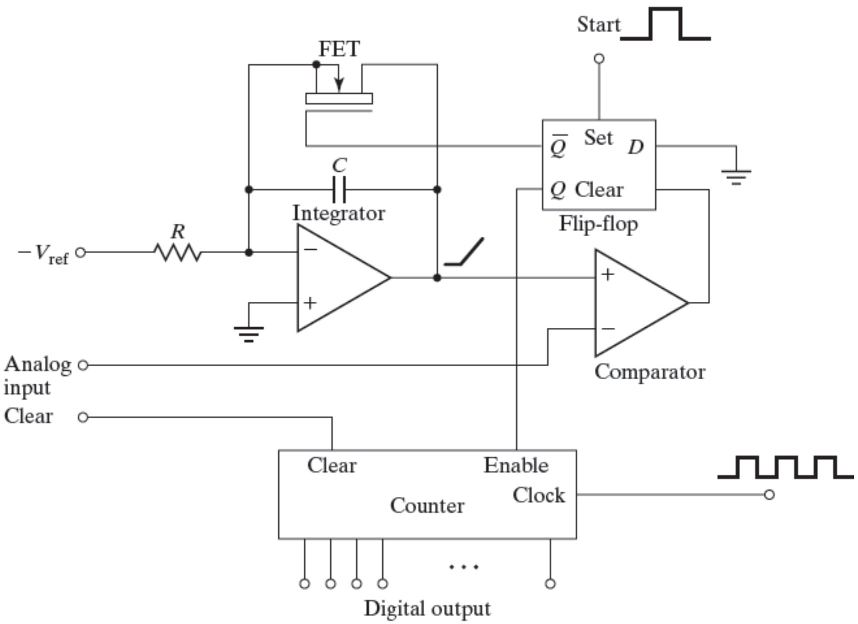
Increment (resolution) = $\frac{1}{2^1} = 0.5\text{V}$

Quantization & encoding: example

- A 4-bit single-slope integrating ADC has an input range of 0 to 10 V. Compute the digital output for an analog input of 6.115V?
- **Solution**:
  - The incremental voltage increase in each clock tick is (10 - 0)/2^4 = 0.625(V) → the integrator voltage at the end of each clock tick: 0.625x1=0.625, 0.625x2 = 1.250, …, 0.625x10=6.250 > 6.115 → stop counter → the output of the converter is 10, which has the value 1010 when expressed in binary.
  - Generalization: $D = int\left\{\frac{V_i - V_{rl}}{V_{ru} - V_{rl}} 2^N\right\} = int\left\{\frac{6.115-0}{10} 2^4\right\} = int(9.78) = 10.$
- A 12-bit A/D converter has an input range of -10 to + 10 V. Find the input resolution error of the converter for the analog input. What should we do if the input voltage is 0.1(V)?
- **Solution**:
  - The input resolution error is $\pm 0.5\left[\frac{10-(-10)}{2^{12}}\right] = \pm 0.00244$ V.
  - If the input voltage is 0.1 V (which has a magnitude at the low end of the input range), the quantization error would then represent ±2.5% of the reading, which is probably not acceptable. The solution to this problem is to amplify the input (with some well-defined gain such as 10) before the signal enters the converter.

Single-slope integrating converter





- A fixed reference voltage is used to charge an integrator at a **constant rate** → the integrator output voltage will then increase **linearly** with time, and is compared continuously with the analog input voltage using a comparator.
- A digital counter is started at the same time that the charging is begun.
- When the integrator voltage exceeds the analog input voltage, the digital clock is stopped → the count is the digital output of the A/D converter.

Example

- A 4-bit single-slope integrating ADC has an input range of 0 to 10 V. Compute the digital output for an analog input of 6.115V?
- **Solution**:
  - The incremental voltage increase in each clock tick is (10 - 0)/2^4 = 0.625(V) → the integrator voltage at the end of each clock tick: 0.625x1=0.625, 0.625x2 = 1.250, …, 0.625x10=6.250 > 6.115 → stop counter → the output of the converter is 10, which has the value 1010 when expressed in binary.
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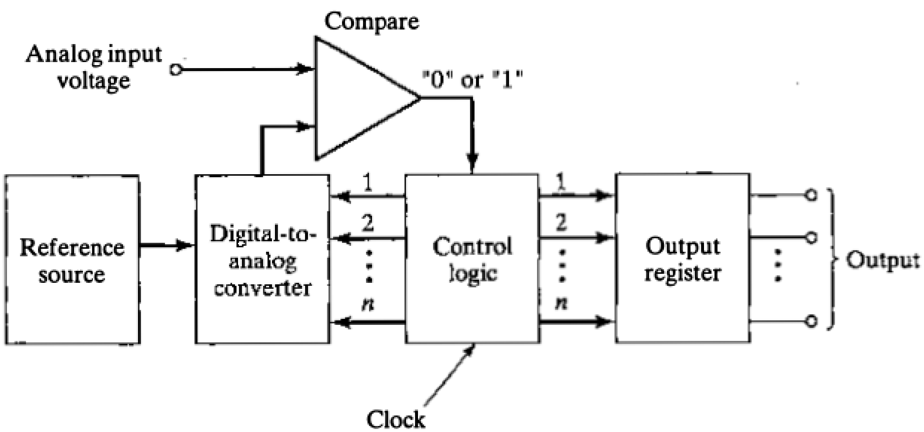
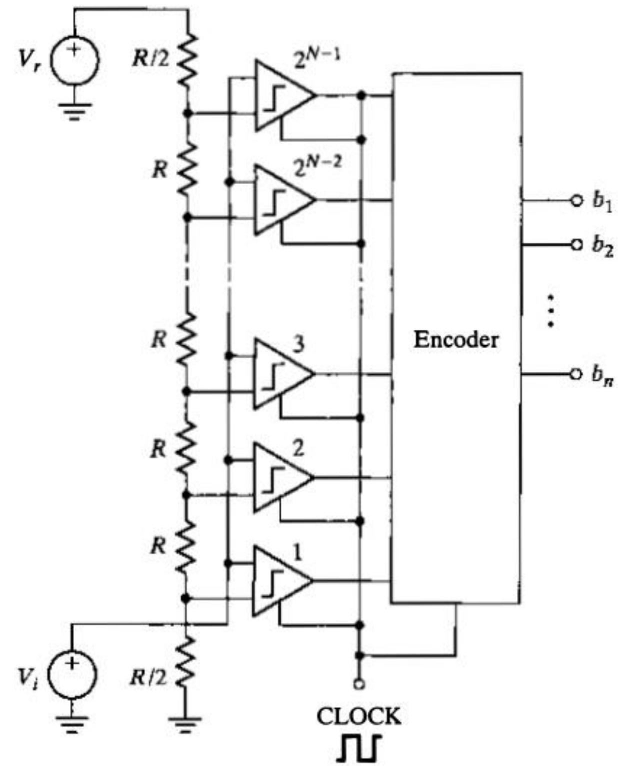
Other common types of ADC

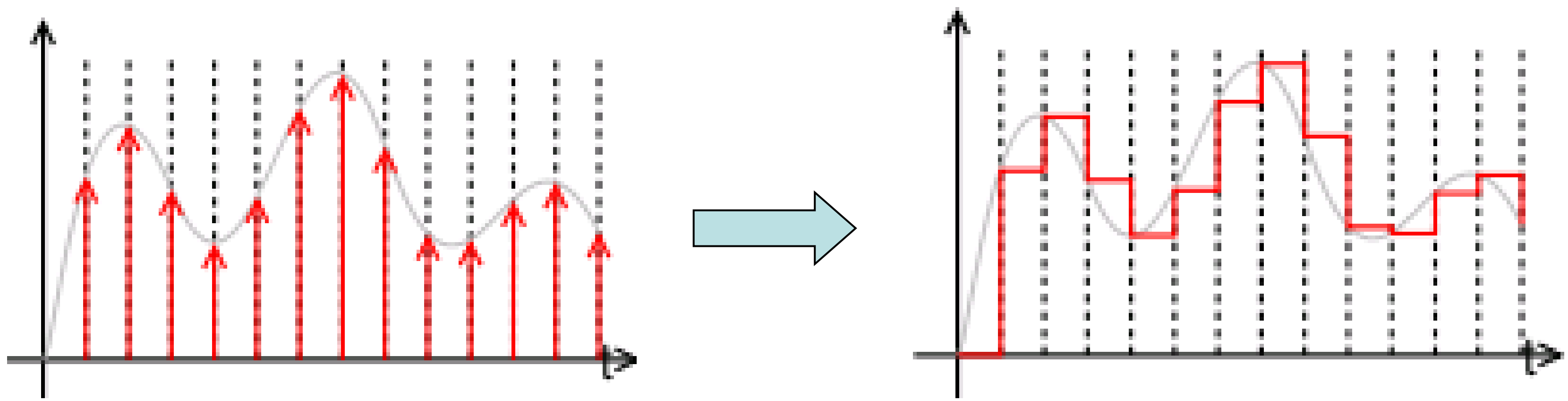


- The most common type is the **successive-approximations converter**, in which a series of known analog voltages are created and compared to the analog input voltage.
- In the first trial, a voltage interval of one-half the input span is compared with the input voltage → determines the MSB.



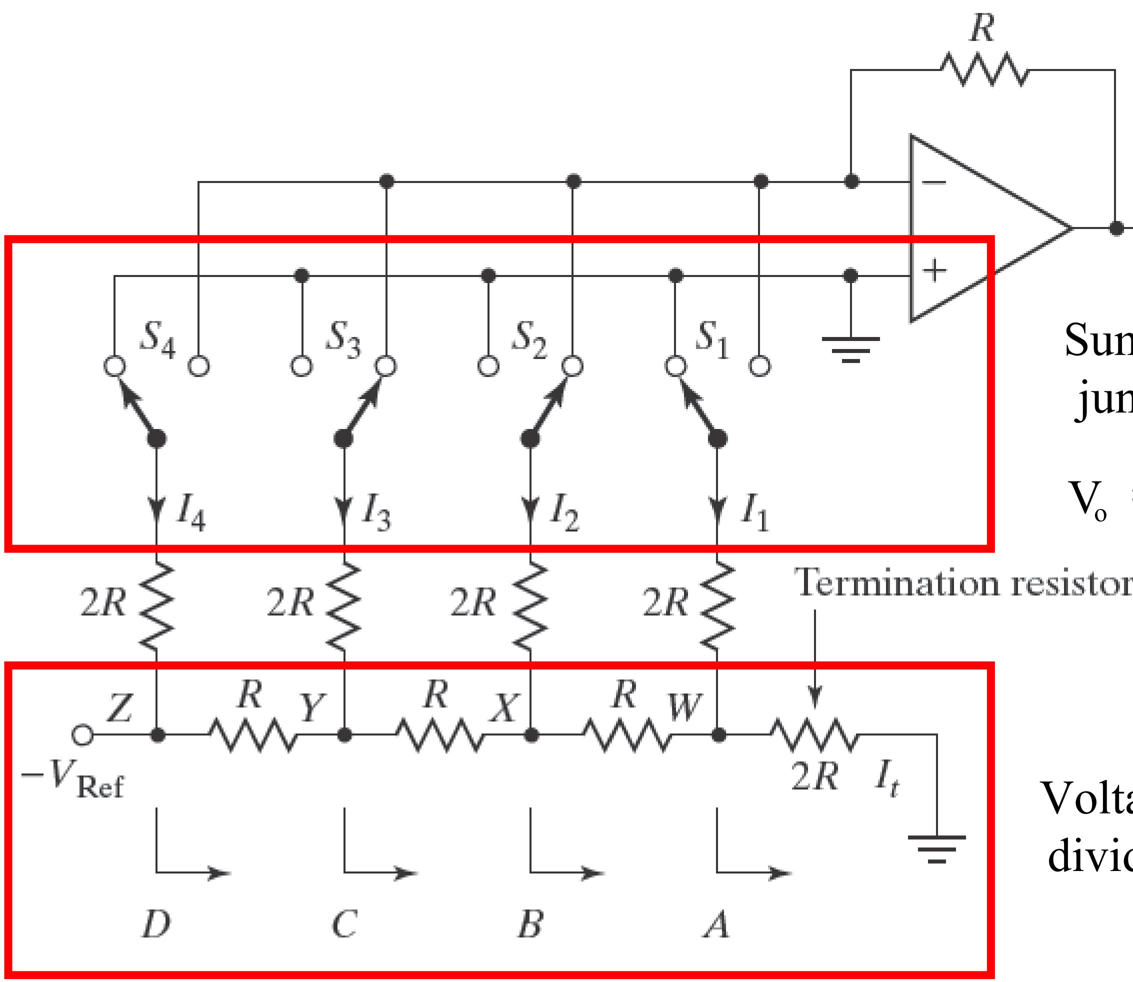
- The fastest type of A/D converter is the **parallel** or **flash** A/D converter where the conversion process is completed in a single step.
- Parallel converters require a large number of components (e.g., $2^N - 1$ comparators).

Digital-to-Analog Conversion (DAC)



- The DAC fundamentally converts finite-precision numbers into a physical quantity, usually an electrical voltage.
  - Normally the output voltage is a linear function of the input number.
  - These numbers are usually updated at uniform sampling intervals and can be thought of as numbers obtained from a sampling process.
- Output is a sequence of piecewise constant values or rectangular pulses → there is an inherent effect of the zero-order hold on the effective frequency response of the DAC resulting in a mild roll-off of gain at the higher frequencies.

A Simple Digital-to-Analog Converter



Summing junction

$$V_o = K(S_4 \cdot 2^3 + S_3 \cdot 2^2 + S_2 \cdot 2^1 + S_1 \cdot 2^0)$$

Voltage divider

- 4-bit DAC: the resulting analog output voltage will be proportional to the digital input number.

Putting all together

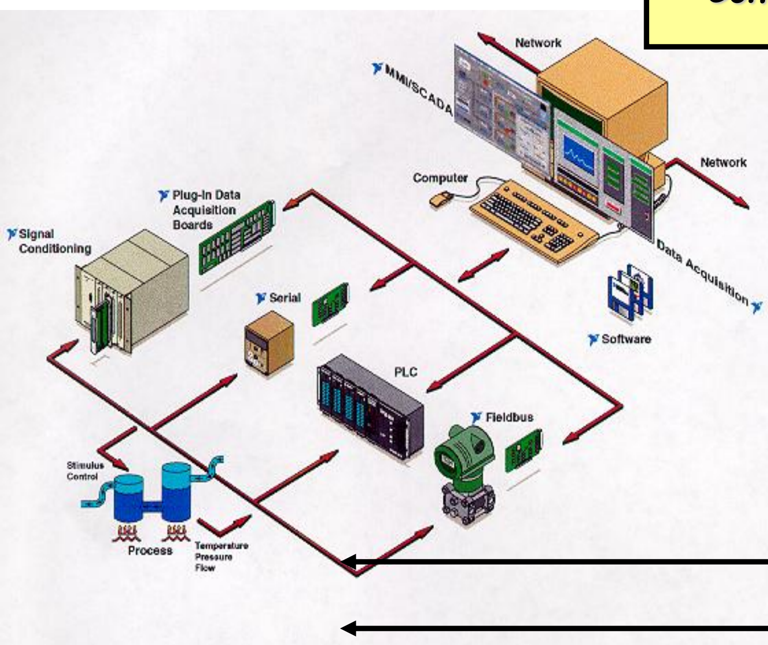
Sensors

Signal Conditioning

A/D Conversion

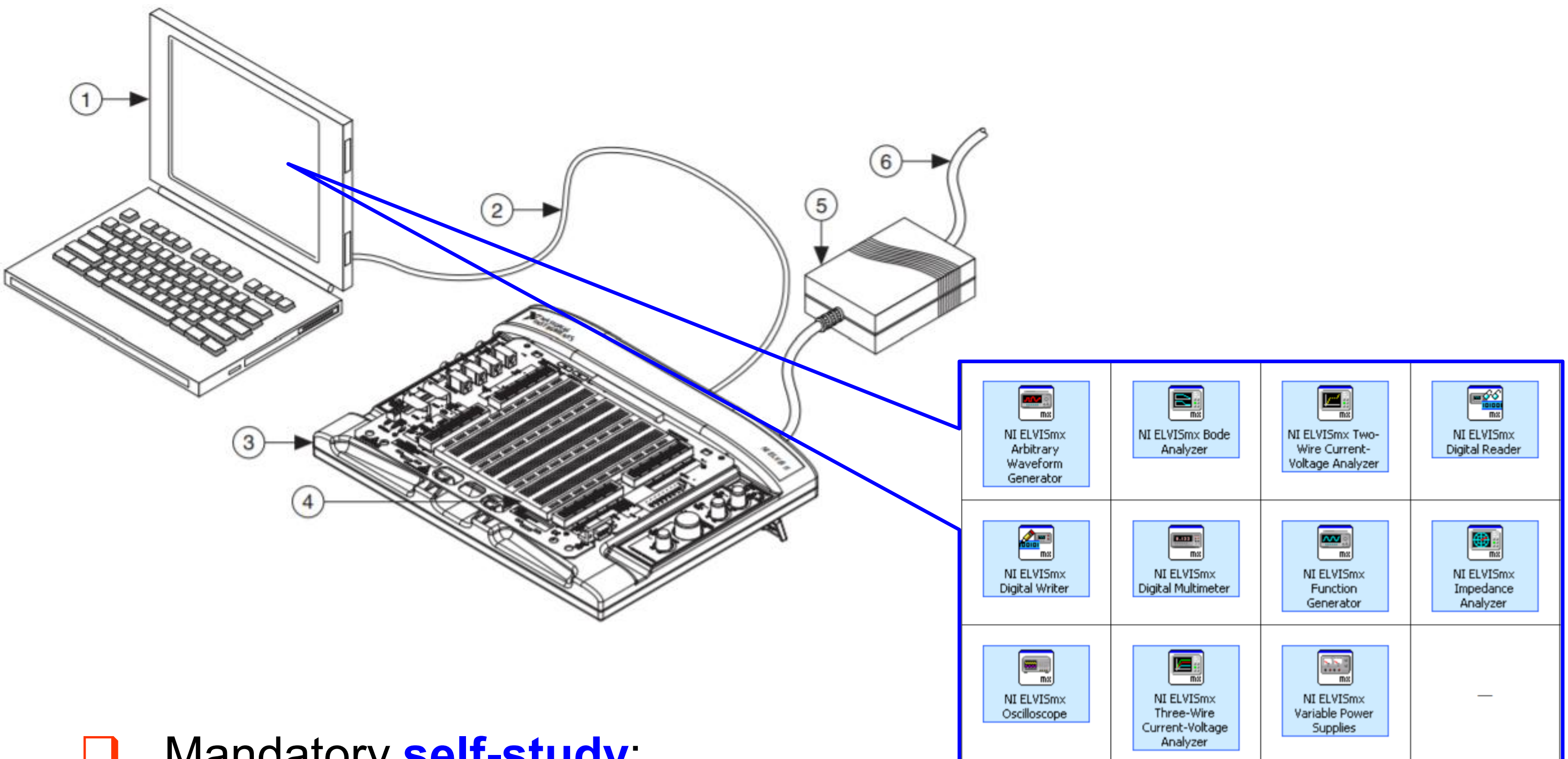
Computer

D/A Conversion



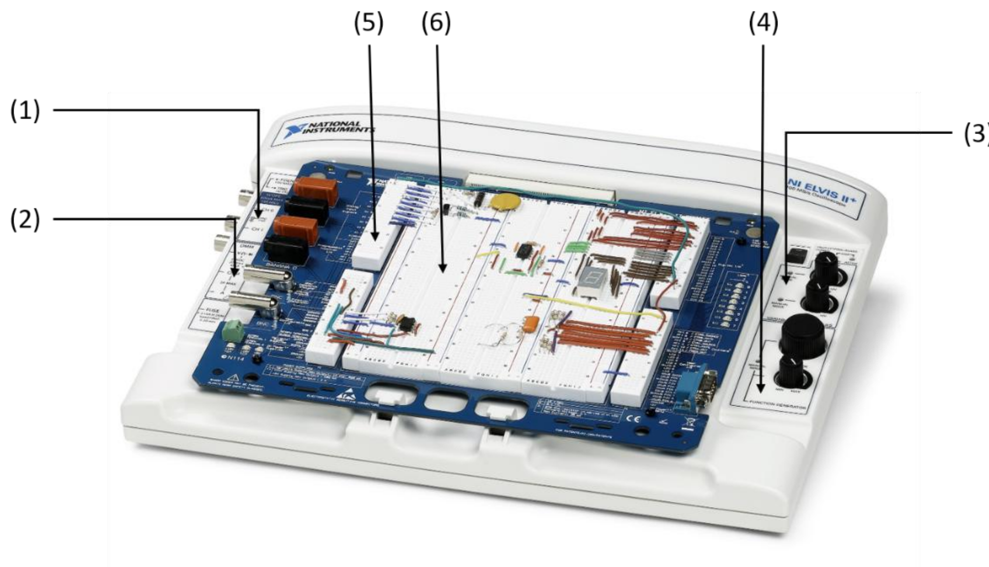
Feedback

The NI ELVIS II+ DAQ system and LabView programming language



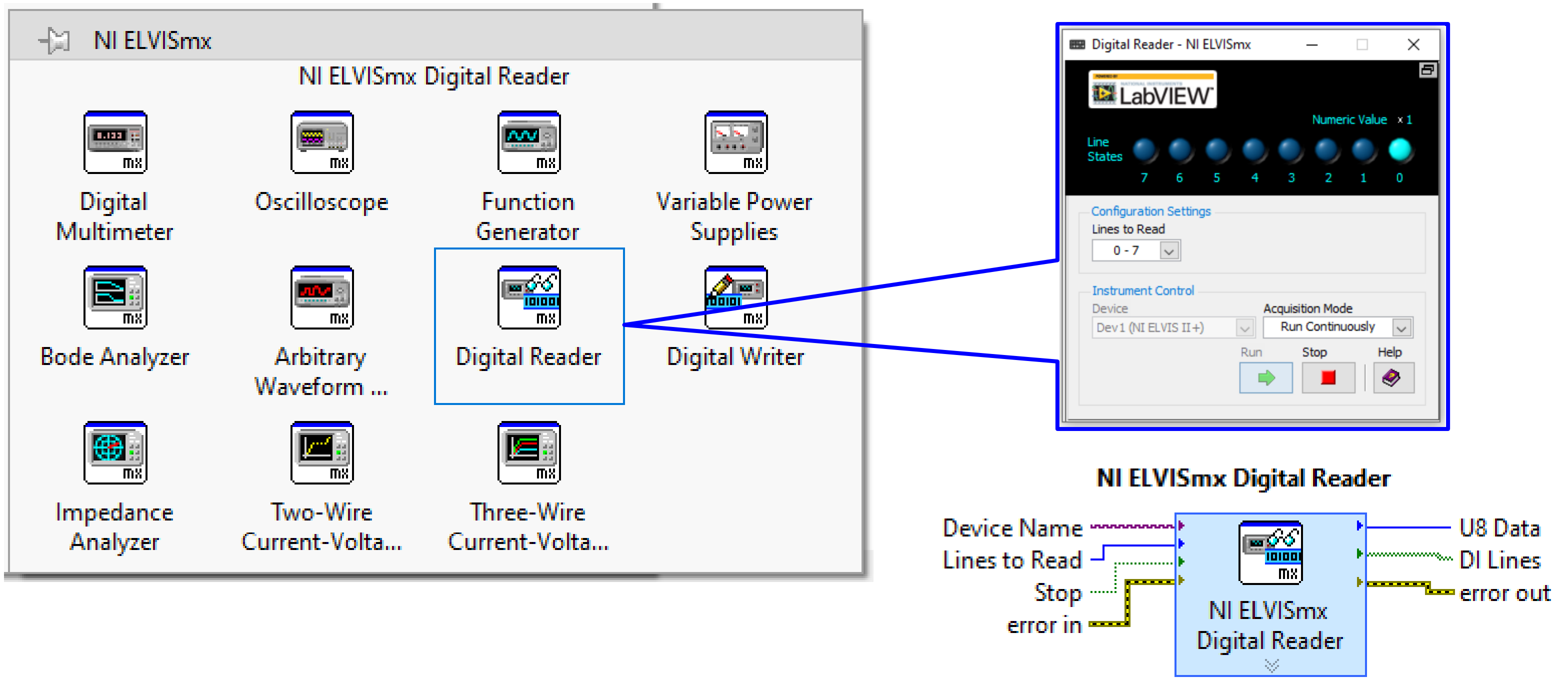
- Mandatory **self-study**:
  - Uploaded materials on the course website
  - https://www.ni.com/getting-started/labview-basics/

The NI ELVIS II+ DAQ system



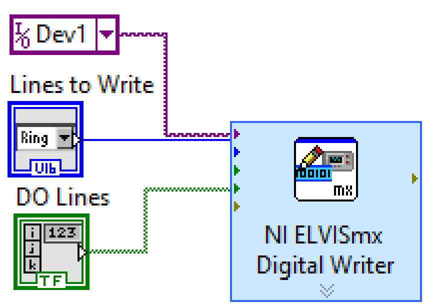
- Some specs/functionalities:
  - 8 differential or 16 single ended analog input channels, 16-bit ADC
  - 24 digital I/O channels
  - 15 PFIs (programmable function interface)
  - Waveform Generator/Analog Output, 16-bit DAC
  - 2 General Purpose Counter/Timers
  - PC interfacing via USB port

Software



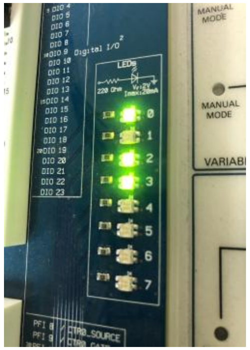
- DAQ functionalities can be programmed via LabView
  - Running on Windows PC
  - The main control GUI is called NI ELVISmx, which provides many Soft Front Panels, e.g., Digital Multimeter (DMM), Bode Analyzer, Oscilloscope, Function Generator, Digital Reader/Writer, …
  - LabView is a (relatively) simple graphical HLL

Examples

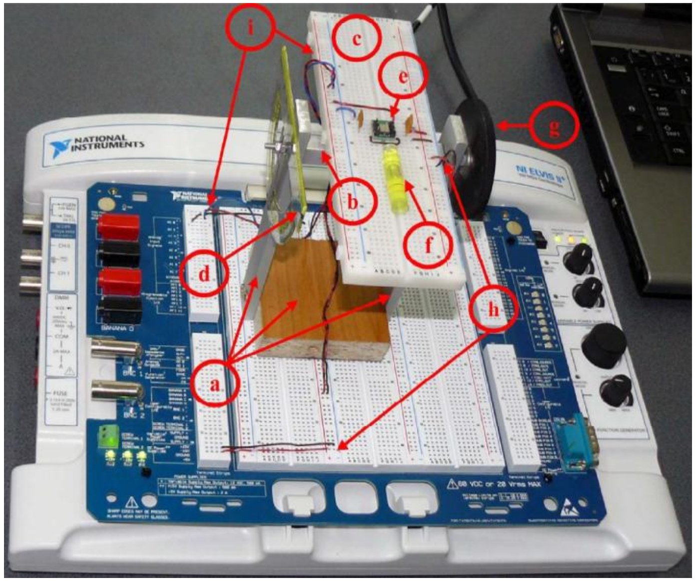


Dev1
Lines to Write
Ring
U16
DO Lines
123
TF
NI ELVISmx Digital Writer

Lines to Write
0 - 7
DO Lines
0



Digital I/O
LEDs
MANUAL MODE



i
c
e
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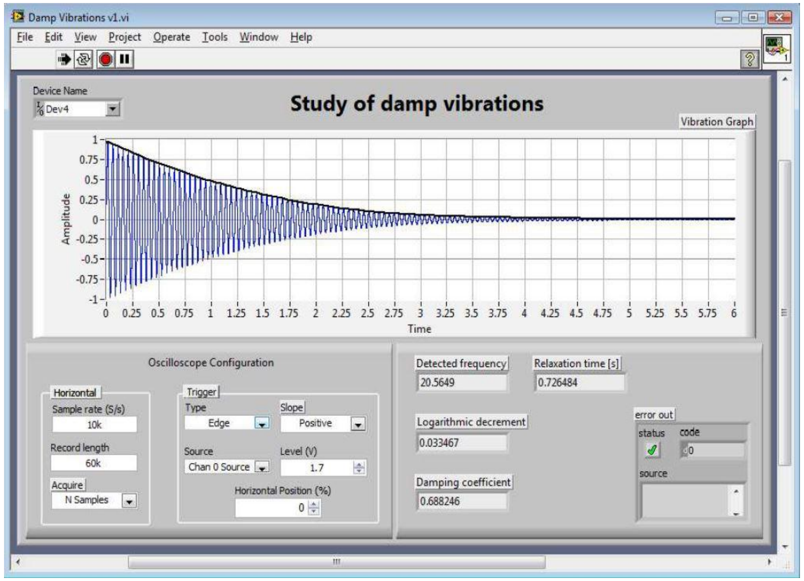
d
h
a
NATIONAL INSTRUMENTS



Damp Vibrations v1.vi
File Edit View Project Operate Tools Window Help
Device Name
Dev4
Study of damp vibrations
Vibration Graph
Amplitude
Time
Oscilloscope Configuration
Horizontal
Sample rate (S/s)
10k
Record length
60k
Acquire
N Samples
Trigger
Type
Edge
Slope
Positive
Source
Chan 0 Source
Level (V)
1.7
Horizontal Position (%)
0
Detected frequency
20.5649
Relaxation time [s]
0.726484
Logarithmic decrement
0.033467
Damping coefficient
0.688246
error out
status
code
0
source

Summary

- Principles of a DAQ board
  - A/D process: sampling, quantization, encoding
  - Common A/DC converters
  - D/A process
  - R/2R ladder DAC
- Introduction to NI ELVISII+ DAQ board
- **Self-study**: NI ELVISII manual + Getting started with LabView
- **Next lecture**: I/O interfacing with computer